AI Shield: the systems I built

Abdelrahman Soliman | Image detection, model evaluation, and recorded provenance

1 Original ML pipeline

EfficientNet + custom VAE
OpenForensics + FaceGate
Custom ViT attention head
Forensic + semantic GLM

2 Detector replacement

New multi-generator dataset
Same-image comparison
Two-stage CF fine-tuning
Second fusion experiment

3 Current implementation

Frozen CF inference
EXIF metadata extraction
C2PA credential checks
Combined FastAPI backend

Current system: fine-tuned Community Forensics + separate provenance evidence.

Forensic layer: three different visual signals

Tiny-GenImage supplied general images; OpenForensics supplied face-manipulation training.

EfficientNet-B0

Learn real/fake artifacts.

Trained on 28,000 images; calibrated scores and cached predictions.

Custom real-only VAE

Test reconstruction error.

Built encoder, decoder, latent sampling, losses, and training loop.

OpenForensics + gate

Detect face manipulation.

Fine-tuned face model; FaceGate uses its score only with a valid face.

FaceGate records applicability, face count, size and confidence before fusion.

What the forensic experiments changed

Calibration, reconstruction testing, and two learned fusion alternatives.

Calibration improved recall

At cutoff 0.5: 75.2% -> 88.6%.
Score adjustment made EfficientNet less conservative.

VAE reconstruction overlapped

Standalone AUROC ~0.470.
Kept as a supporting feature in the original fusion, not its main detector.

Phase 1 held-out split	AUROC
Calibrated EfficientNet	0.9517
Logistic fusion (GLM)	0.9542
Small neural fusion (MLP)	**0.9561**

Small gains from forensic fusion; next, test image-region relationships.

Semantic layer: my custom attention head

Pretrained ViT patch features + a trainable PyTorch head.

ViT patch tokens > **Multi-head attention** > **MLP refinement** > **Attention pooling**

What I implemented

Queries, keys and values; patch interactions;
refinement and learned pooling.
856,578 trainable parameters.

What the test showed

AUROC 0.9256; recall 90.63%.
Caught many generated images,
but also flagged 797 / 3,500 real images.

Useful complementary signal; tested with the forensic features in Phase 3.

Phase 2 notebook

Final fusion: combine evidence from the same image

A logistic-regression GLM learned how much weight to give each feature.

11 forensic features

EfficientNet, VAE, face score, face presence and applicability

6 semantic features

ViT attention model outputs and derived score features

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17-feature GLM

Learned weights for a single real/fake score

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Final prediction

Threshold chosen on validation

Fusion splits: 4,200 train | 1,400 validation | 1,400 test.

Phase 3 results: full GLM fusion

All rows below use the same 1,400-image Tiny-GenImage fusion test.

Model	AUROC	Accuracy
Calibrated EfficientNet	0.9558	90.29%
Semantic attention	0.9303	85.93%
Naive average	0.9763	91.71%
Forensic-only GLM	0.9578	89.71%
Full forensic + semantic GLM	**0.9819**	**94.07%**

54 more generated images caught, with 1 extra false positive vs. EfficientNet.

New data: multiple generators + real photographs

5,124 images total: 2,562 real and 2,562 generated.

Generated: Qwen-Image-Bench

FLUX.2 max / GLM-Image / Seedream 5
Nano Banana 2 / Qwen-Image-2.0-pro

GPT Image 2 reserved for a separate test.

Real: Open Images V7

Real photographs provide the comparison class and reveal false alarms.

Split	Images
Training	2,990
Validation	640
Ordinary test	640
Held-out generator	**854**

The old pipeline struggled on the newer images

Re-ran the saved original model without retraining it.

Original benchmark

Tiny-GenImage fusion test

AUROC 0.9819

Strong original result.

New ordinary test

Dataset v1: 640 images

AUROC ~0.606

Weak transfer.

Held-out generator

GPT Image 2 evaluation

AUROC ~0.599

Weak transfer here too.

Test a stronger detector on these same images: Community Forensics.

Comparison notebook

Adapted Community Forensics in two stages

Started from the published OwensLab/commfor-model-384 checkpoint.

Pretrained baseline	No project fine-tuning	**0.8780**
Stage 1: classifier head	3 epochs; backbone fixed	**0.9804**
Stage 2: final blocks	3 epochs; last 2 blocks + head	**0.9870**

Right column: validation AUROC. Final test images did not choose the checkpoint.

Selected the Stage-2 checkpoint, then evaluated on the two reserved test sets.

CF training notebook

CF alone beat the old system and the new fusion

Same final images for every row. Higher AUROC means better ranking.

Detector	Ordinary test	GPT Image 2
Old AI Shield full fusion	~0.606	~0.599
Original CF	0.8860	0.9555
CF + older forensic branches	0.9854	0.9936
Fine-tuned CF alone	**0.9884**	**0.9946**

Fine-tuned CF accuracy: 95.16% ordinary test; 96.60% held-out evaluation.

Selected CF alone. The second fusion did not improve either AUROC.

What stays in the current pipeline

The original models remain documented experiments; CF is the selected detector.

Selected

Fine-tuned CF: one visual AI score.

Same checkpoint, preprocessing, and validation max-F1 cutoff ~0.634.

Six reference images reproduce the saved predictions through the API.

Archived / paused

EfficientNet, VAE and face branch.
Custom semantic attention head.
Original GLM and newer fusion.

These explain the research journey; they are not active in the new backend.

Freeze = hold the chosen version fixed while building the rest of AI Shield.

EXIF: read what the original file records

Implemented with ExifTool; this component needs no model training.

Extracted fields

Camera / device information
Editing software
Embedded creation / modification dates
GPS, only when explicitly requested

Illustrative metadata report

Camera: not recorded
Software: Adobe Photoshop
Creation time: not recorded
Origin: unknown

The original upload is inspected before the detector resizes or crops the image.

No metadata means unknown. An editing-software tag does not establish AI generation.

C2PA: verify signed creation and editing records

Uses the C2PA SDK to check available embedded credentials.

1 Signature

Has the signed record been altered?

Check the cryptographic signature.

2 Image binding

Does that record still match this image?

Detect changes to the signed image content.

3 Signer trust

Do we trust the signer under our policy?

A valid signature alone does not establish trust.

Reports absent, untrusted and failed states separately. Verification currently runs offline.

Signed history is additional evidence; it does not prove that a scene really happened.

One upload now runs detection and provenance

FastAPI connects the frozen model and evidence extraction locally.

Original image upload

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Frozen CF detector

Prepared pixels -> AI score

ExifTool + C2PA

Original bytes -> evidence

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Combined report

Detection result
Metadata details
Credential checks

38 automated tests passed; 6 saved images also matched through the upload API.

Built locally; the new website is the next product step

The existing Hugging Face demo still serves the older fusion system.

Implemented

Selected and verified CF detector
EXIF extraction + C2PA verification
Combined upload API
Automated tests and saved artifacts

Next

JPEG / resize robustness tests
Artwork, renders and edited-photo tests
Commit / publish the backend
Build and deploy the upload website

Later: image/claim consistency checks and short-video support.